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GATE 2009, ECE Question Number 36

Abstract
This project implements the logic equation given in a GATE 2009 ECE question using AVR-GCC and physical hardware (ATmega328P/Arduino). Logical analysis determines the value of variable Z, and the expression is validated using an LED output.

1. Question and Simplification

- Given: $X = 1$
- Expression: $(X+Z)(Y+(Z+Y))(X+Z(X+Y)) = 1$
- Simplifies to: $(1)(Y+Z)(1+Z(1+Y)) = 1$
- Solving: $Z(1+Y) = 0 \Rightarrow Z = 0$

Correct Answer: (D) Z = 0

2. Components

Component	Qty
ATmega328P / Arduino UNO	1
Push Buttons (X, Y, Z)	3
LED for Output	1
Resistors (220Ω for LED, 10kΩ for pull-downs)	4
Breadboard	1
Jumper Wires	10
Laptop with AVR-GCC	1

Table 1: List of components used

3. Setup and Connections

- Connect buttons to PD0, PD1, PD2 for inputs X, Y, Z.
- Connect LED to PB0 with 220Ω resistor.
- All buttons should use 10kΩ pull-down resistors.
- Common GND and VCC lines shared across all components.

4. Logic Summary

- Inputs: X, Y, Z (button-controlled)
- Expression: $(X+Z)(Y+(Z+Y))(X+Z(X+Y))$
- With X = 1, simplify to: $(Y+Z)(1+Z(1+Y))$
- Final condition: LED ON if entire expression evaluates to 1

5. Pin Mapping

- **X** – PD0 (Input)
- **Y** – PD1 (Input)
- **Z** – PD2 (Input)
- **LED** – PB0 (Output)

6. Hardware Code – AVR GCC (C)

```c
#define F_CPU 1000000UL
#include <avr/io.h>
#include <util/delay.h>

int main(void)
{
    DDRD = 0x00; // Inputs: PD0 - X, PD1 - Y, PD2 - Z
    DDRB = 0xFF; // Output: PB0 - LED
    PORTD = 0x00;

    while (1)
    {
        unsigned char X = (PIND & (1 << PD0)) >> PD0;
        unsigned char Y = (PIND & (1 << PD1)) >> PD1;
        unsigned char Z = (PIND & (1 << PD2)) >> PD2;

        unsigned char part1 = (X || Z);
        unsigned char part2 = (Y || (Z || Y));
        unsigned char part3 = (X || (Z && (X || Y)));

        unsigned char output = part1 && part2 && part3;

        if (output)
            PORTB |= (1 << PB0); // LED ON
        else
            PORTB &= ~(1 << PB0); // LED OFF

        _delay_ms(200);
    }
    return 0;
}
```

7. Analysis

7.1 Truth Table (sample)

X	Y	Z	Output
1	0	0	1
1	1	0	1
1	0	1	0
1	1	1	0

Table 2: Logic validation for Z

7.2 Circuit Diagram

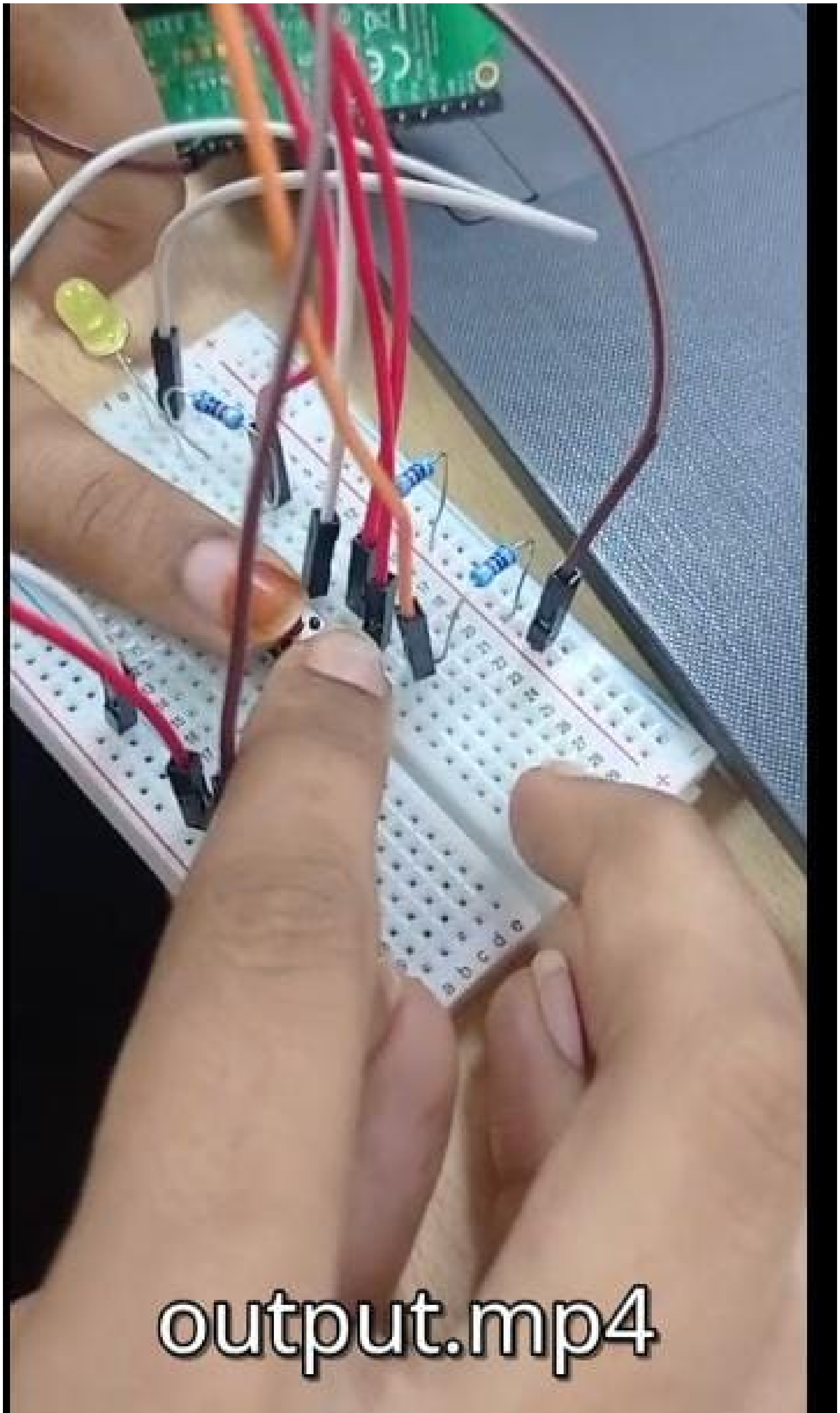

Figure: Practical Circuit Using AVR

8. Conclusion

This project demonstrated evaluation of a GATE logic question both theoretically and practically using AVR-GCC. The LED output confirms that the expression is valid only when $Z = 0$, verifying the analytical result.